

Optimizing Scattered Intensity vs. Absorption

- Beer's Law: $I/I_0 = e^{-\mu t}$ where I is the scattered beam, I_0 is the incident beam intensity, μ is the absorption coefficient and t is the greatest thickness of the crystal. Note: if μ is in mm^{-1} t is in mm.
- The optimal value of $\frac{I}{I_0}$ is 0.5:

$$e^{-\mu t} = 0.5$$

$$-\mu t = \ln(0.5)$$

$$t = 0.693/\mu$$

Stout and Jensen, X-ray Structure Determination: A Practical Guide